

Practical No. 04

ON

SSCS4020 - Cryptography & Network Security

TITLE: To implement the Vigenère Cipher algorithm for encrypting and decrypting a plaintext message using a keyword.

**BACHLEOR OF SCIENCE IN INFORMATION TECHNOLOGY
(HONOURS.)**

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Pr1:

```
def encrypt(text, key):  
    result = ""  
    key = key.upper()  
    key_index = 0  
  
    for ch in text.upper():  
        if ch.isalpha():  
            shift = ord(key[key_index % len(key)]) - ord('A')  
            result += chr((ord(ch) - ord('A') + shift) % 26 + ord('A'))  
            key_index += 1  
        else:  
            result += ch  
  
    return result
```

```
def decrypt(text, key):  
    result = ""  
    key = key.upper()  
    key_index = 0  
  
    for ch in text.upper():  
        if ch.isalpha():  
            shift = ord(key[key_index % len(key)]) - ord('A')  
            result += chr((ord(ch) - ord('A') - shift) % 26 + ord('A'))  
            key_index += 1  
        else:  
            result += ch  
  
    return result
```

```
plaintext = input("Enter Plaintext: ")  
keyword = input("Enter Keyword: ")  
  
ciphertext = encrypt(plaintext, keyword)  
print("Encrypted Text :", ciphertext)  
  
original = decrypt(ciphertext, keyword)  
print("Decrypted Text :", original)
```

```
● devanpatel@Devans-MacBook-Air Assignments % /usr/local
Cyberspace/Assignments/Practical 5/pr1.py"
Enter Plaintext: INSTITUTE0FCOMPUTERAPPLICATION
Enter Keyword: PPSU
Encrypted Text : XCKNXIMNTDXWDBHOITJUEEDCRPLCDC
Decrypted Text : INSTITUTE0FCOMPUTERAPPLICATION
```

Pr2.

```
def encrypt(message, key):
    result = ""
    key = key.upper()
    key_index = 0

    for ch in message.upper():
        if ch.isalpha():
            shift = ord(key[key_index % len(key)]) - ord('A')
            result += chr((ord(ch) - ord('A') + shift) % 26 + ord('A'))
            key_index += 1
        else:
            result += ch

    return result

def decrypt(cipher, key):
    result = ""
    key = key.upper()
    key_index = 0

    for ch in cipher.upper():
        if ch.isalpha():
            shift = ord(key[key_index % len(key)]) - ord('A')
            result += chr((ord(ch) - ord('A') - shift) % 26 + ord('A'))
            key_index += 1
        else:
            result += ch

    return result

print("----- Sender -----")
message = input("Enter Confidential Message: ")
secret_key = input("Enter Shared Secret Key: ")
```

```
encrypted = encrypt(message, secret_key)
print("\nCipher Text Sent:", encrypted)

print("\n----- Receiver -----")
received_key = input("Enter Shared Secret Key: ")

decrypted = decrypt(encrypted, received_key)
print("Original Message:", decrypted)
```

```
● devanpatel@Devans-MacBook-Air Assignments % /usr/local/bin
Cyberspace/Assignments/Practical 5/pr2.py"
----- Sender -----
Enter Confidential Message: NETWORK SECURITY
Enter Shared Secret Key: CYBER

Cipher Text Sent: PCUAFTI TITWPJXP

----- Receiver -----
Enter Shared Secret Key: CYBER
Original Message: NETWORK SECURITY
```